Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Calculate $3 + 2 \times 4$.

2. Calculate 2^0, 2^3 and 2^{-2}.

3. A count of 80 increases by 5%. Give the new count.

4. For $f(x) = 3x - 2$, calculate $f(0)$ and $f(4)$.

5. How many integers are there from 3 to 10 inclusive?

6. A loop `for k in range(4)` takes which values of k ?

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. In 90 seconds, explain the difference between index, term and number of terms. Give a counterexample to a rule guessed from three values.

Investigate before formalising

In pairs, compare two collection schemes: 20 kg in the first week, then 5 kg more each week; 20 kg then a 20% increase each week. Conjecture which collects the most in total in six weeks, then justify. Allow 10 min of research and 5 min of comparison.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.